



UNIVERSITY OF NORTH CAROLINA
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MITIGATING RUSSIAN CYBERATTACKS ON THE INTERNET OF THINGS

Consumer: Dave Luber



Figure 1

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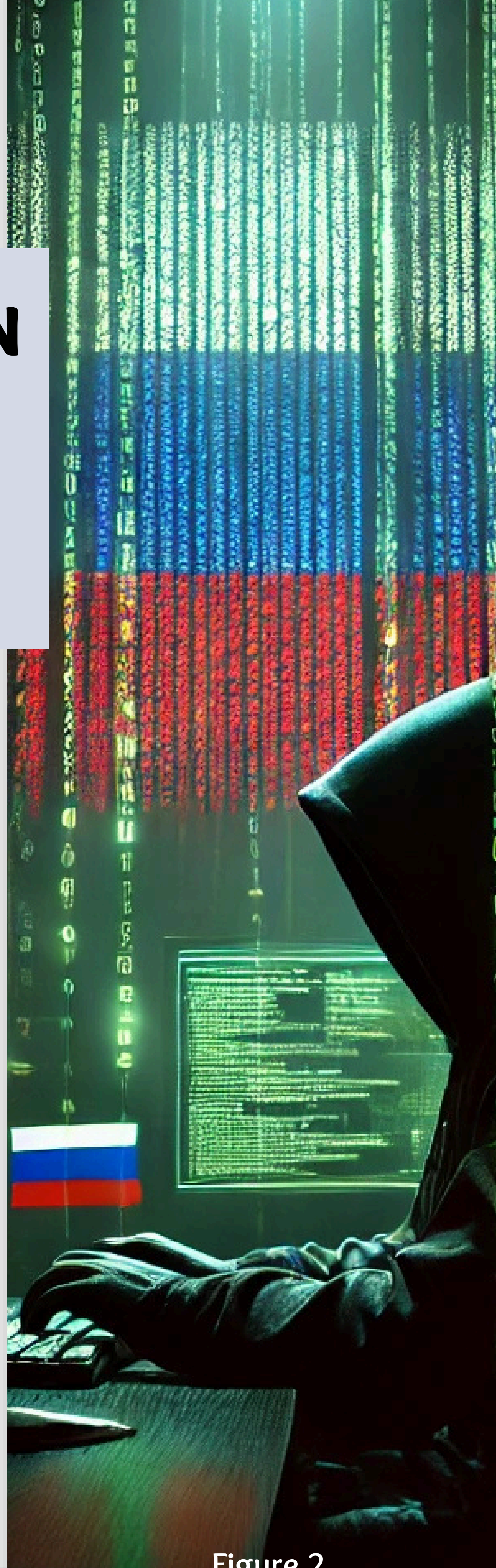


Figure 2

EXECUTIVE SUMMARY

Russia has grown bolder with their cyberattacks across the United States, Ukraine, and many NATO nations. Their ability to attack the Internet of Things, which almost all American citizens use, is not only a privacy threat but a national security threat. This policy brief begins with a summary of Russian attacks thus far, methods used by Russian hackers to attack the IoT, and concludes with our policy recommendations to reduce and prevent future attacks. These recommendations are the establishment of existing National Institute of Standards and Technology guidelines as regulatory practices for IoT devices, the reinstatement of the United States Computer Emergency Readiness Team to provide recovery support post-attacks, and the improvement of cyber resilience on the individual level through cyber safety education.

INTRODUCTION

Russian cyber actors have been able to infiltrate and carry out attacks on the Internet of Things (IoT) through scanning tools, firmware vulnerabilities, and botnet exploitation. The United States has over 3.6 billion devices connected to the IoT.¹ It is widely known that devices are vulnerable to cyber attacks due to their lack of ensured universal security. The large number and diversity of devices connected to the IoT require varying regulations and protocols.² Since these devices are connected, the hacking of one device could compromise data from another.³ Data leaks are one of the many vulnerabilities of the IoT, risking the privacy and security of the US.⁴ Over the past two decades, Russia has posed an ever-increasing global cyber threat to the United States. With the increase in IoT devices, Russian cyber actors have had a greater surface of attack.

This issue has become more pressing with Russian actors taking drastic actions to derail US democracy. In 2016, Russian hackers compromised the Hilary Clinton campaign through data leaks in an attempt to increase Donald Trump's odds of becoming president.⁵



Figure 3

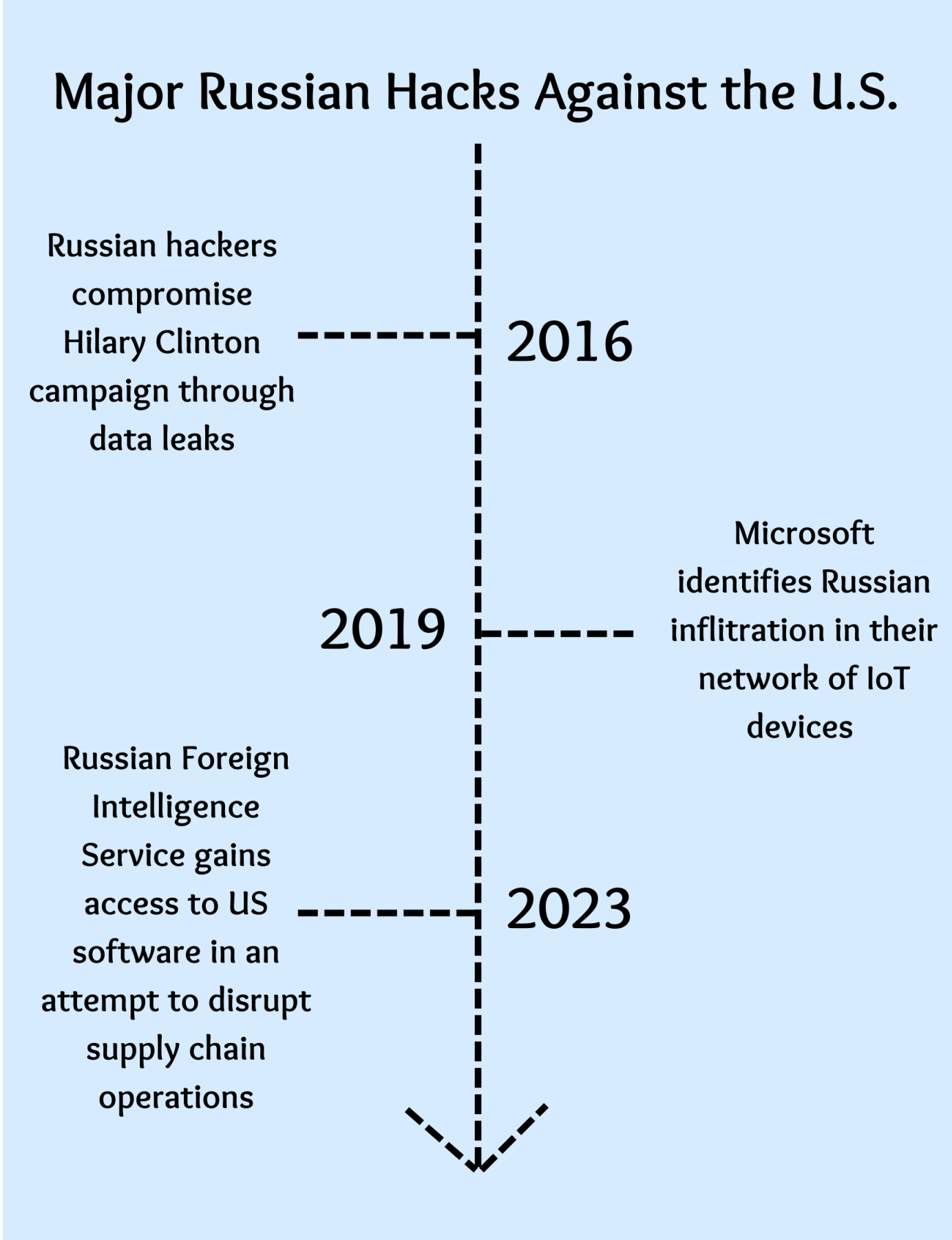
In 2019, Microsoft found Russian infiltration in their networks hacking into numerous IoT devices.⁶ In 2023, the Russian Foreign Intelligence Service gained access to US software in an attempt to disrupt supply chain operations.⁷ With the recent events surrounding the Russia-Ukraine war, Russia is lacking economic and political resources, increasing its reliance on espionage and cyber attacks. This puts the United States IoT in a vulnerable position.

Understanding these attacks and others is of utmost importance to United States national security. Providing defense against Russia's use of scanning tools, firmware vulnerabilities, and botnet exploitation is crucial to protecting and ensuring the future of our privacy and democracy. Based on our analysis of the various cyberattacks and vulnerabilities in the IoT, we have developed three recommendations to mitigate Russian cyberattacks on the Internet of Things. These recommendations require numerous actions that must be taken nationally and individually to protect the United States and its citizens.

METHODOLOGY



We gathered information on cyberattacks from the FBI’s statement on Russian inference in the 2016 presidential election and the NSA’s on Russian attempts to disrupt supply chain operations to establish a need for policy implementation. We then used sources such as the American Cyber Defense Agency (CISA), the National Security Agency (NSA), and a Joint Cybersecurity Advisory to explain the Russian methods utilized to attack the IoT. Finally, we researched existing policies on the federal, state, and individual levels to combat cybersecurity attacks. Sources such as the National Institute of Standards and Technology, CISA, and the Institute for Defense and Business were used to shape our recommendations.



Timeline of major Russian hacks against the Internet of Things, spanning from 2016 to present day

RESULTS & CONCLUSIONS



As the Internet of Things (IoT) has continued to develop and expand its surface over the years, Russian cyber actors have been more drawn to exploiting the IoT. Russia has always emphasized the importance of technologies and their ability to infiltrate the US cyber world, which has been made clear as we analyze their recent activity. Most of the recent activity carried out by Russian hackers to target the IoT can be seen under these three methods: scanning tools, exploiting firmware vulnerabilities, and using Botnets.

SCANNING TOOLS:

Scanning tools are a broader method used by cyber actors to find devices that are connected through the Internet of Things. IoT devices are known to collect and transmit large data sets that are often unencrypted, making it a prime target for cyber attacks. Unit 29155 is a group of Russian cyber actors employed under the General Staff of the Armed Forces of the Russian Federation (GRU) and are responsible for attacks against networks for espionage and sabotage.⁸ Some of the public tools available to them include “Acunetix, Amass, Droopescan, eScan, and JoomScan.” Specifically, the Acunetix tool is used to gather information on vulnerabilities within the device network so hackers can infiltrate the systems, which is another method cyber actors use.⁹ In 2019, Microsoft became aware of Russian hackers infiltrating their encrypted computer networks. They discovered that the Russian cyber actors got into printers, video decoders, and other

devices with firmware vulnerabilities that they found connected through the IoT. Once they penetrated these systems, they used simple network scans to find vulnerable devices and gain access to privileged accounts and high-value data.¹⁰



Figure 4

Emblem of the Main Directorate of the General Staff of the Armed Forces of the Russian Federation (GRU)

FIRMWARE VULNERABILITIES:

IoT devices are often outdated or have insufficient protection, making them an easy target for Russian hackers. Russian cyber actors can exploit known vulnerabilities in systems to gain access or alter system operations. In 2023, the NSA put out a statement to warn that Russian Foreign Intelligence Service (SVR) actors are taking advantage of a known vulnerability by the name CVE-2023-42793 (a software development program). This allows actors to have initial access to the TeamCity servers (internet-connected global servers) and then access other devices through the internet-connected network.¹¹ TeamCity servers give SVR actors access to source code and the ability to corrupt software compilation and interrupt supply chain operations.¹² Many victims fell to this hack, “including companies that

“THE FOUR MAIN TYPES OF BREACHING A SECURITY SYSTEM ARE (ATTEMPTS OF) UNAUTHORIZED ACCESS TO A SYSTEM OR DATA, THE UNAUTHORIZED USE OF SYSTEMS THAT PROCESS DATA, CHANGES TO SYSTEMS WITHOUT THE OWNERS’ CONSENT OR ¹³ DISRUPTION OF SERVICES”

provide software for billing, medical devices, customer care, employee monitoring, financial management” and more.¹⁴ The SVR has used its initial access into the TeamCity CVE to further its prerogative by opening additional backdoors and taking steps to ensure long-term access to the network.¹⁵

BOTNETS:

IoT botnets are a network of devices connected to the IoT that are infected by malware and controlled by cyber actors. Botnets are known to be used to launch DDoS attacks (distributed denial-of-service) on targets to interrupt their system services and operations. Botnets gain power in numbers, meaning that the more devices that make up a botnet the more powerful it is. Generally, a single command-and-control server controls botnets and is connected to all the “bots” (infected devices).¹⁶ In 2021, Fortinet (a cybersecurity company) first observed Moobot. Moobot is a malware that is based on the Mirai botnet (malware used to launch the largest DDoS attack ever). Russian hackers targeted routers using the Moobot malware to install and repurpose their botnet to serve as a global cyber espionage platform. This botnet was used to target U.S. intelligence agencies and unnamed military organizations.¹⁷

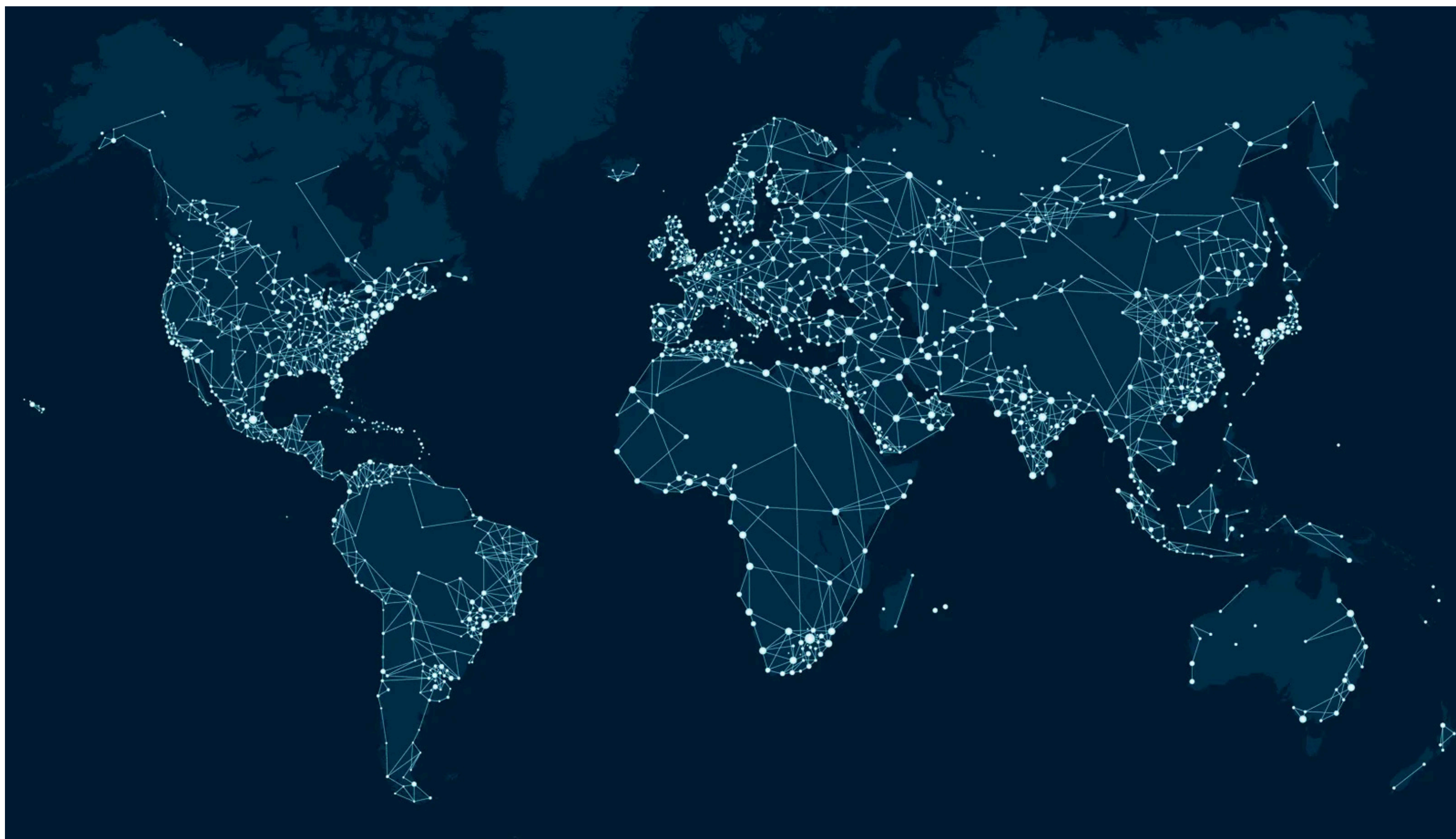


Figure 5

IoT botnets don't only pose a challenge to U.S. security, but also to other botnets. Botnets are designed to grow and compete with one another to gain control over infected devices. This creates a "worm war" where botnet malware families use their tools to infect as many devices as possible while simultaneously fighting off other botnets!¹⁸ This shows IoT botnets' aggression and ability to infect large amounts of interconnected devices.

CONCLUSION:

Russian cyber actors have been utilizing methods to infiltrate US devices interconnected through the Internet of Things. Access to the IoT allows hackers to deepen their foothold in US networks, further jeopardizing the privacy and security of information and business operations. The longer these methods lack a US response, the greater the threat posed by Russian cyber actors.

RECOMMENDATIONS



Russian cyberattacks on the Internet of Things cannot be stopped or minimized with just one solution. Due to Russia's complex and ever-growing network of hacker groups and constant technological development, fighting against malicious cyber attacks grows more difficult each day. That being said, we have a number of recommendations to be implemented on a national policy level and a local level that will help mitigate this threat. These recommendations focus specifically on minimizing the risk of cyberattacks rather than improving U.S. and Russia relations as a whole.

REGULATION OF IOT DEVICES:

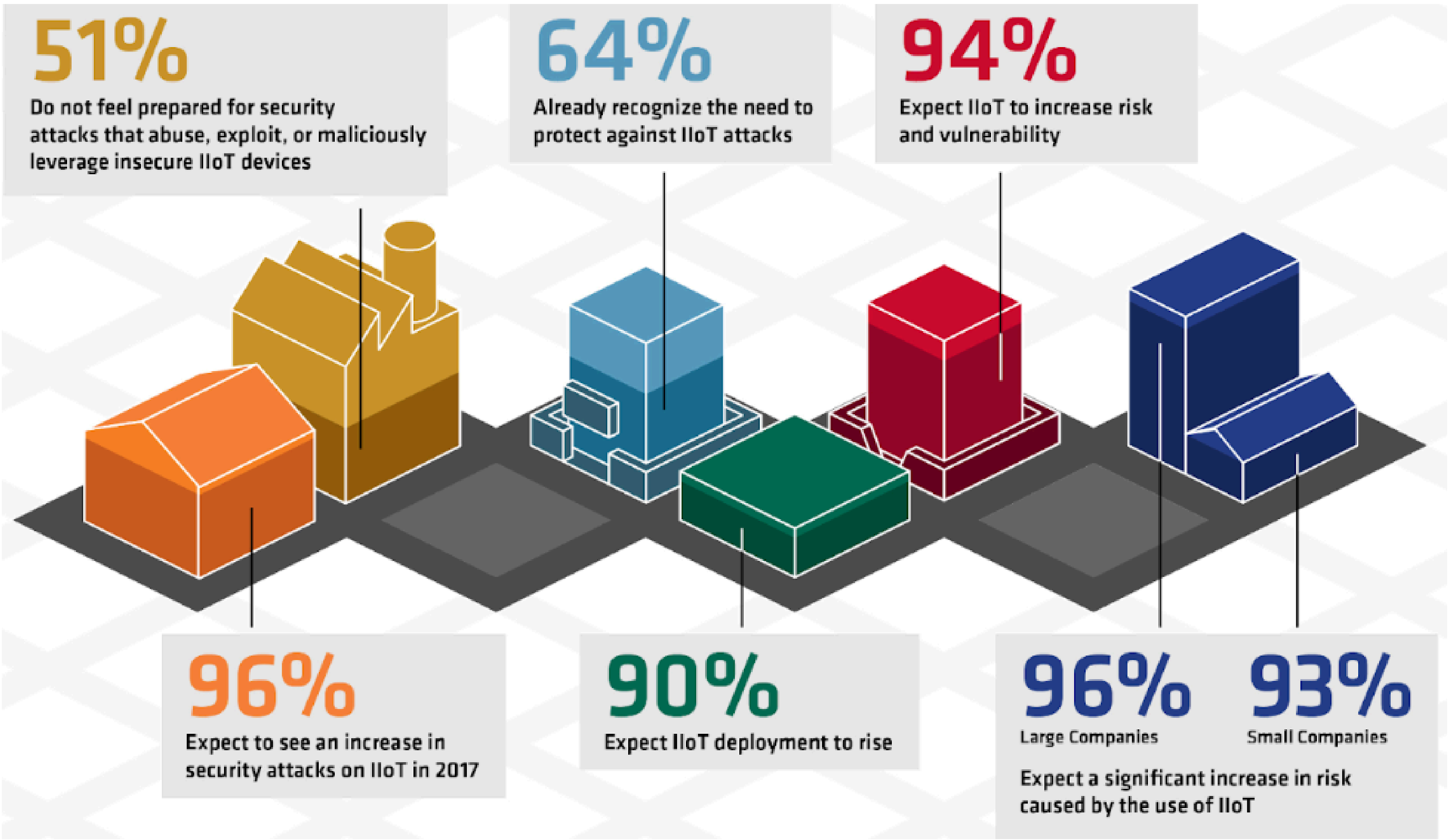
The Internet of Things has been a force in the cyber world for less than two decades and there is limited policy regulating these devices.¹⁹ This policy gap reveals the need for more federal intervention regarding the Internet of Things and the world of cyber as a whole as technology continues to progress. Policy that does exist seems to make suggestions and recommendations rather than hold IoT devices to a specific and hard standard. To strengthen the security of IoT devices and decrease their vulnerability to attacks, we recommend utilizing existing guidelines from the National Institute of Standards and Technology (NIST) and establishing them as required regulatory practices. The NIST’s Special Publication 800-213 details numerous standards for IoT standards that should be maintained for maximum

security against malicious cyber activity.²⁰ These recommendations, along with input from CISA, should be made into federal policy to ensure that all IoT devices are held up to a high standard of security from their inception through years into use.

REINSTATEMENT OF US-CERT:

When cyberattacks do occur, it is essential to salvage the data stolen and damage inflicted as soon as possible. To achieve this, the US-CERT (United States Computer Emergency Readiness Team) has played a key role in the past. In 2023, US-CERT was retired with ICS-CERT (Industrial Control Systems Cyber Emergency Response Team) and integrated into CISA (Cybersecurity and Infrastructure Security Agency).²¹ We recommend reviving US-CERT as a specific emergency-oriented branch of CISA for the fastest possible response times to Russian cyberattacks on the IoT. This

Figure 6



Corporations understand both the value and risk associated with the IoT, but many still feel unprepared to secure their devices.

branch should be housed under CISA's cybersecurity or infrastructure security division.²² In the past, US-CERT has maintained a 24/7 Secure Operations Center, provided malware recovery support for government agencies, managed malicious cyber activity, and collaborated with corporations, federal agencies, and regional programs to work on cybersecurity.²³ We recommend reinstating US-CERT to improve response times to Russian hacking and malicious cyber behavior.

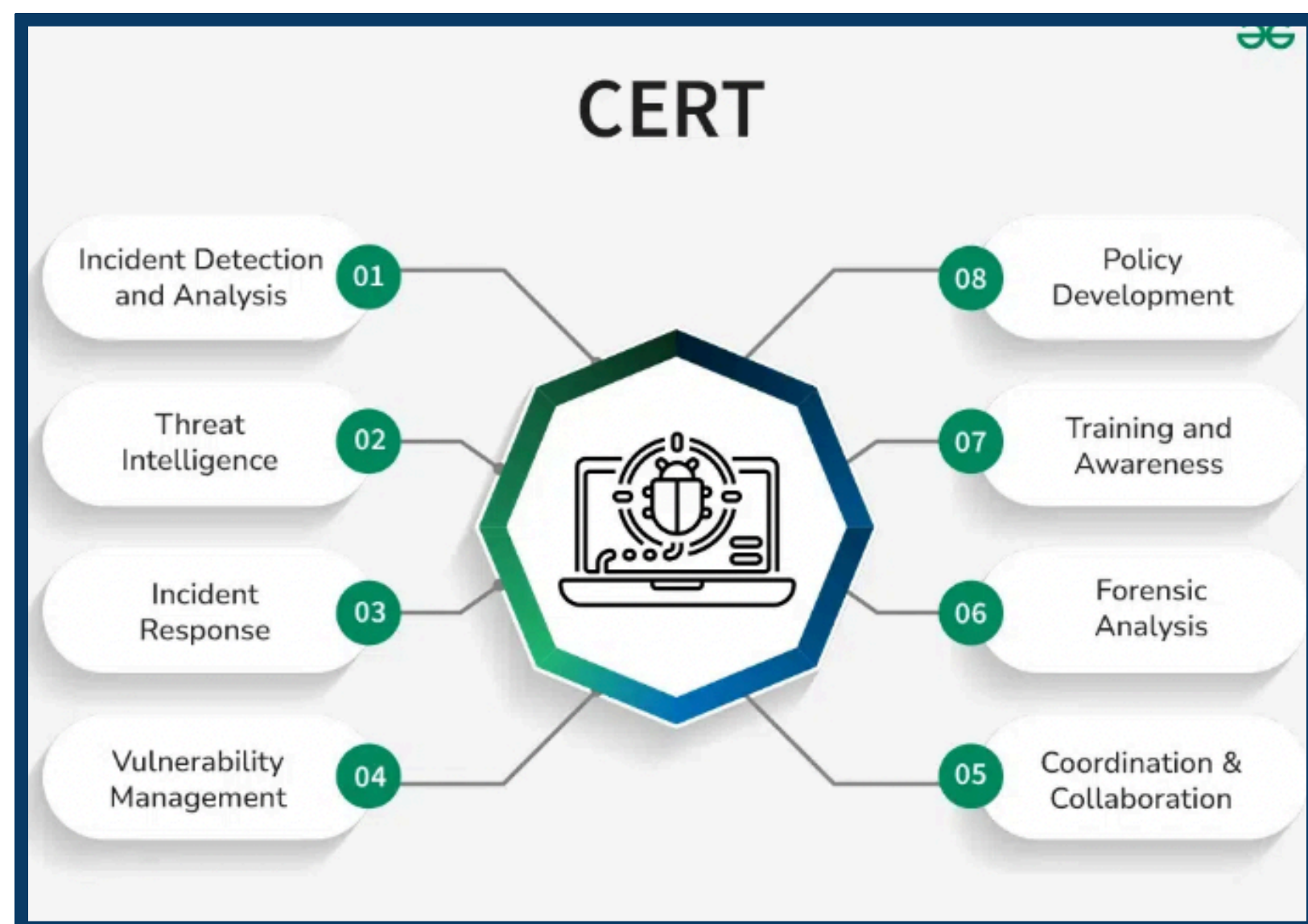


Figure 7

SMALL-SCALE SOLUTIONS:

Aside from policy changes and national-level actions against Russian cyberattacks, citizens and corporations can also protect themselves on an individual level by improving their cyber resilience. This can be done by following CISA advisories which emphasize the importance of implementing cybersecurity performance goals, fixing common network misconfigurations, and immediately reporting malicious activity to CISA, among other strategies.²⁴ To further disrupt cyber attacks to the IoT, organizations should avoid generic or repeated passwords for multiple devices and should constantly update and patch software. This will make it more difficult for one hack to grant access to an entire system or household's network of things.²⁵ While these measures are not foolproof, these actions bolster cybersecurity on a local level and allow for individuals and

organizations to take some level of responsibility against malicious cyber activities. We recommend implementing policies that require corporations to educate employees on these cyber safety measures.

CONCLUSION:

To decrease the frequency and severity of Russian cyberattacks, the U.S. can take several steps at varying levels of policy. Regulating IoT devices based on NIST's standards, reinstating US-CERT as an emergency response team to cyberattacks, and encouraging protection and education on a small scale for corporations and individuals would all be effective options toward mitigating risk.

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